Shaine Wei

+1 (607) 319-9094 | wshu07@gmail.com | linkedin.com/in/shaine-wei | shoewhen.github.io

Education

Duke University, Pratt School of Engineering Durham, NC

B.S.E. in Biomedical & Mechanical Engineering, Minor in Mathematics *May 2029 (Expected)*

- BME Design Fellow | Dean's List | GPA: 3.87/4.00
- **Relevant Coursework:** Quantitative Physiology & Biostatistical Applications, Molecular Biology, Organic Chemistry I, Linear Algebra, Advanced Multivariable Calculus, Ordinary and Partial Differential Equations, Data Structures & Algorithms, Engineering Innovation, Fundamentals of Electrical and Computer Engineering

Experience

Undergraduate Researcher Oct. 2026 – Present

Bridgeman Lab, Duke University *Durham, NC*

- Developing a debris-mitigation system for a robot-mounted surgical laser used in autonomous tumor ablation
- Diagnosing how ablation debris contaminates the camera's protective window and degrades imaging accuracy
- Designing a CAD model of the laser end-effector for a physical mock-up to test debris-mitigation concepts

Undergraduate Research Assistant Sept. 2026 – Present

EBC Laboratory, Duke University *Durham, NC*

- Analyzing human biological samples for air pollution exposure biomarkers using mass spectrometry and chromatography
- Contributing to research linking environmental toxicant exposure to adverse human health outcomes

Co-Director & Tech Team Lead Sept. 2025 – Present

HackDuke, Duke University *Durham, NC*

- Lead HackDuke's executive team in organizing Duke University's annual hackathon, coordinating logistics for 200+ attendees
- Lead development of HackDuke's full-stack event platform using React, Python, and PostgreSQL, overseeing website reliability
- Mentor new developers through technical workshops, code reviews, and hands-on guidance in full-stack web development

Technical Communications Assistant Sep. 2025 – Present

Mechanical Engineering & Materials Science, Duke University *Durham, NC*

- Lead production of the bi-semester undergraduate newsletter in collaboration with faculty and research groups
- Translate technical research topics into accessible written content for a broad student audience
- Support departmental communications and event documentation

Projects

EV Brake Assembly Redesign (Duke Electric Vehicles) | *Mechanical Design, Braking Systems* Sept. 2026 – Present

- Diagnosing brake drag in which the disc contacts the pads even when the brakes are disengaged
- Redesigning the brake assembly to restore pad-to-disc clearance and minimize parasitic drag on the drivetrain

Rube Goldberg Machine (EGR121L) | *Arduino, Sensors, Mechanical Design* Apr. 2026

- Built a multi-stage electromechanical Rube Goldberg course on a 15-person team using Arduino-controlled sensors
- Led design and integration of an Arduino-controlled elevator that returned the ball to the start for a second run
- Coordinated timing across subteams to deliver reliable, cohesive performance of the full system

Sunflower Shelling Machine (EGR121L) | *SolidWorks, 3D Printing* Feb. 2026

- Designed and fabricated a human-operated shelling machine achieving 2nd highest performance score in a 30+ student class
- Engineered a fully 3D-printed kernel/shell separation system with removable sifting tray and integrated collection chamber
- Modeled assemblies in SolidWorks and integrated a gear-driven dual-roller compression mechanism to minimize kernel damage
- Conducted quantitative testing of seed geometry and crushing force to inform channel spacing and roller cut dimensions

Yeast Genome Cloning Project (BIO201L) | *Molecular Biology* Jan. – Apr. 2026

- Extracted genomic DNA from yeast cells and performed enzymatic digestion using RNase and restriction enzymes
- Constructed recombinant plasmids via ligation; applied antibiotic selection and screening to isolate successful clones
- Performed PCR and gel electrophoresis to amplify and analyze DNA fragments

Technical Skills

Programming: Python, Java, R, HTML/CSS, JavaScript, React, Git
CAD & Fabrication: SolidWorks, Fusion 360, Inventor, Onshape, 3D printing, laser cutting, woodworking
Lab Techniques: PCR, gel electrophoresis, immunohistochemistry, microscopy, mass spectrometry, chromatography
Data & Tools: R/Python data analysis, Excel, LaTeX, Arduino, microcontrollers
Languages: English (fluent), Mandarin Chinese (conversational)

Activities & Organizations

American Society of Mechanical Engineers (ASME) | Biomedical Engineering Society (BMES) | DTech | Duke Electric Vehicles | HackDuke | Society of Women Engineers (SWE)